

Assignment Day 7 - Pointers

1. Print Address

```
#include <iostream>
using namespace std;

int main() {
    int num = 10;
    int *ptr = &num;

    cout << "Address = " << ptr;
    return 0;
}
```

Input: No Input

Output: Address = (memory address)

2. Swap Using Pointers

```
#include <iostream>
using namespace std;

void swapNum(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

int main() {
    int x = 10, y = 20;

    swapNum(&x, &y);

    cout << x << " " << y;
    return 0;
}
```

Input: 10 20

Output: 20 10

3. Array Sum Using Pointers

```
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {10,20,30,40,50};
    int *ptr = arr;
    int sum = 0;

    for(int i=0;i<5;i++)
        sum += *(ptr+i);

    cout << "Sum = " << sum;
    return 0;
}
```

Input: No Input

Output: Sum = 150

4. Largest Element Using Pointers

```

#include <iostream>
using namespace std;

int main() {
    int arr[5] = {10,25,8,40,15};
    int *ptr = arr;
    int max = *ptr;

    for(int i=1;i<5;i++) {
        if(*(ptr+i) > max)
            max = *(ptr+i);
    }

    cout << "Largest = " << max;
    return 0;
}

```

Input: No Input

Output: Largest = 40

5. Traverse Array Using Pointers

```

#include <iostream>
using namespace std;

int main() {
    int arr[5] = {1,2,3,4,5};
    int *ptr = arr;

    for(int i=0;i<5;i++)
        cout << *(ptr+i) << " ";

    return 0;
}

```

Input: No Input

Output: 1 2 3 4 5

6. Reverse Array Using Pointers

```

#include <iostream>
using namespace std;

int main() {
    int arr[5] = {1,2,3,4,5};
    int *ptr = arr;

    for(int i=4;i>=0;i--)
        cout << *(ptr+i) << " ";

    return 0;
}

```

Input: No Input

Output: 5 4 3 2 1